PES UNIVERSITY

SOFTWARE ENGINEERING

CUSTOMIZABLE SUBSCRIPTION BOX SCHEDULER

Name:shivanand

SRN:PES2UG24CS472

Section:5H

1. Architectural Choice:
The Subscription Management System is divided into Client Application, Subscription Management, Payment Gateway, Fulfillment & Shipping, and Data Management components connected through ports and interfaces.

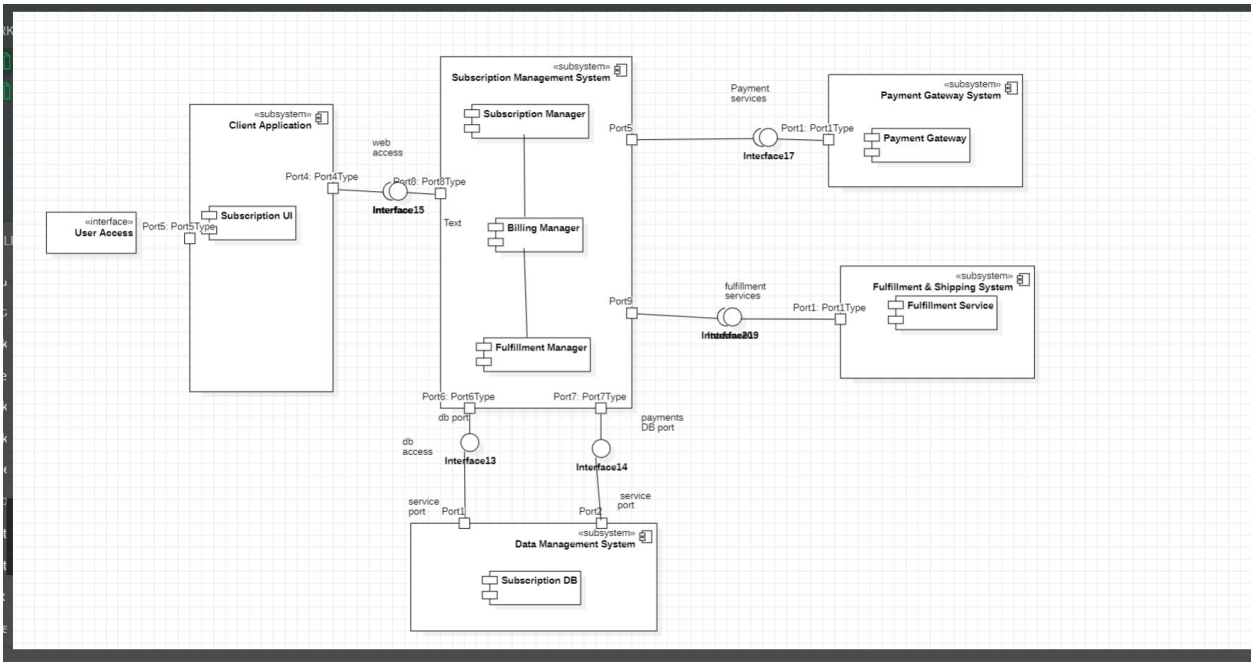
2. Two Reasons:

- **Modularity:** Each component handles a separate function, making development and maintenance easier.
- **Clear communication:** Components communicate through defined interfaces such as Web Access, Payment Services, Fulfillment Services, and DB Access.

3. Security Advantage:
Defined interfaces provide controlled access between components and reduce unnecessary exposure.

4. Performance Benefit:
Independent components can process operations separately, reducing unnecessary processing and improving responsiveness.

5. Component Diagram:


«subsystem»
Subscription Management System
Subscription Manager
Billing Manager
Fulfillment Manager
«subsystem»
Client Application
Subscription UI
«interface»
User Access
Port5: Port5Type
Port4: Port4Type
Port8: Port8Type
web
access
Interface15
Text
Port5
Payment
services
Interface17
Port1: Port1Type
«subsystem»
Payment Gateway System
Payment Gateway
Port9
fulfillment
services
Port1: Port1Type
«subsystem»
Fulfillment & Shipping System
Fulfillment Service
Port6: Port6Type
Port7: Port7Type
db port
payments
DB port
db
access
Interface13
Interface14
service
port
Port1
Port2
service
port
«subsystem»
Data Management System
Subscription DB